

Reg No: MGSE1A0027Name: MADHAV KANISHKAB

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: AIT312

Course Name: RECOMMENDATION SYSTEM

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

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| 1 | What is the central concept behind collaborative recommendation? | (3) |
| 2 | Describe the approaches for implicit and explicit ratings in collaboration system. | (3) |
| 3 | Explain knowledge based recommendation. | (3) |
| 4 | Define Constraint Satisfaction Problem (CSP). | (3) |
| 5 | What are the limitations of hybridization strategies? | (3) |
| 6 | What are the two dimensions of hybrid recommendation system? | (3) |
| 7 | Differentiate confidence and trust. | (3) |
| 8 | Explain A/B testing. | (3) |
| 9 | What are the different methods in group attack profile detection? | (3) |
| 10 | What is probe attack? | (3) |

PART B*Answer one question from each module, each carries 14 marks.***Module I**

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| 11 | a) Elaborate on the user-based nearest neighbour recommendation system with an example. | (8) |
| | b) Discuss the methods that tackle the issues of data sparsity and the cold start problem. | (6) |

OR

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| 12 | a) Provide a detailed discussion on content based recommendation system. | (14) |
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Module II

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| 13 | a) How defaults are utilized in constraint based recommendation? | (7) |
| | b) Explain QuickXPlain algorithm. | (7) |

OR

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| 14 | a) Explain critiquing algorithms in detail. | (14) |
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Module III

- 15 a) Explain monolithic hybridization designs in detail. (14)

OR

- 16 a) How does parallelized hybridization work? Explain in detail. (7)
b) Explain about different pipelined hybridization design. (7)

Module IV

- 17 a) Discuss about the various evaluation paradigms in recommendation. (7)
b) Illustrate the design issues in offline recommender evaluation with a case study. (7)

OR

- 18 a) How the offline recommendation system can be evaluated using accuracy metrics. (7)
b) Discuss about the general goals of evaluation design. (7)

Module V

- 19 a) How to detect attacks on recommender system? (7)
b) What are the different strategies to design a robust recommendation system? (7)

OR

- 20 a) What are the different types of attacks on recommender system? (14)
